Omkar Joshi

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Education

North Carolina State University — Raleigh, NC
PhD & Masters in Computer Science — *August 2023 – Present*

- Advised by Dr. Munindar P. Singh
- Cumulative GPA: 4.0/4.0
- Relevant Coursework: Artificial Intelligence, Natural Language Processing, Social Computing and Decentralized AI, LLMs in Security, Controllable AI

COEP Tech — Pune, India
Bachelor of Technology in Computer Science, Minor in Financial Engineering — *Aug. 2016 – May 2020*

- Cumulative GPA: 8.65/10.0
- Undergraduate Thesis: Modelling Option Pricing using Local Volatility Model

Selected Publications [See website for full list]

Norma: Modular Norm-aware Deliberation for LLM-based Agents — NAACL 2027
Omkar Joshi, Fardin Saad, Munindar P. Singh — *Under Review*

- Developed Norma, a modular LLM agent architecture for norm-aware deliberation, separating goal-directed reasoning from normative evaluation.
- Evaluated Norma on 50 scientific collaboration scenarios, reducing norm-violating actions from 23% to 4% and improving goal success from 78.4% to 97.3% under norm-compatible goals, while maintaining 84.6% goal success under goal–norm tension.

Ahoy: LLMs Enacting Multiagent Interaction Protocols [link] — EMAS 2026
Omkar Joshi, Munindar P. Singh, Amit K. Chopra — *Paphos, Cyprus*

- Developed an LLM-based architecture for programming-free enactment of multiagent interaction protocols
- Ahoy achieved 100% success across six evaluation tasks with no protocol violations, compared to 0% success for the Claude-4.5 baseline.

ReFeedGraph: Graph-Grounded Reusable Feedback for Debate Improvement — EMNLP 2026
Omkar Joshi, Priya Pitre*, Timon Ziegenbein, Henning Wachsmuth.* — *Budapest, Hungary*

- Developed ReFeedGraph, a graph-based LLM feedback system evaluated across 3 debate datasets
- Improved weaker-side speeches by up to 0.491 points and produced 59.5% winner-flip rates.

ArgAnalysis35K - A Large Scale Dataset for Argument Quality Analysis [link] — ACL 2023
Omkar Joshi, Priya Pitre*, Dr. Y.V. Haribhakta* — *Toronto, Canada*

- Developed ArgAnalysis35K, a 34.9K-pair parliamentary debate dataset with a relevance model and instance-level annotator-reliability scoring
- Evaluated BERT-based quality prediction, achieving 0.54 Pearson and 0.55 Spearman correlation.

Experience

Full Stack Developer, Equities Tech — June 2020 – July 2022
Credit Suisse Services AG — *Pune, India*

- Designed, developed and deployed a new service to integrate Stock Borrow Agreement data into existing database architecture post new financial legislation being passed in Hong Kong in 2021.
- Migrated 200+ active, interdependent jobs to new servers using Control-M.

Featured Project [See Github/Website for full list]

- LLM Argumentation-based Link Prediction [Code]
  Built a machine learning project that uses LLM-driven argumentation signals from social media interactions to predict future user links and improve over traditional link-prediction baselines.

Technical Skills

Languages: Python, C/C++, , SQL (MySQL), JavaScript, HTML/CSS, R
Frameworks: Supabase, Angular, Node.js, PyTorch, Microsoft Agent Framework
Research Interests: Multiagent Systems, LLMs, Norms, Protocols, Argumentation